PROBABILITY AND STATISTICS I

# Assignment 2

Sample spaces, events, and classical probability

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Name: ..... ID: .....

Group: ..... Date: .....

**Before you start.** Three lines per question, every time: the sample space with its size $N$, the event as a subset with its size $n$, then $P = n/N$. A bare fraction with no $S$ and no event earns part marks at best. Leave answers as exact fractions in lowest terms unless the question asks otherwise, and check that every answer sits between 0 and 1.

**1)** A die is rolled. Find the probability that an even number is obtained.

**2)** Two coins are tossed. Find the probability that two heads are obtained.

**3)** A die is rolled and a coin is tossed. Find the probability that the die shows an odd number and the coin shows a head.

**4)** A card is drawn at random from a deck of cards. Find the probability of getting the 3 of diamonds.

**5)** A card is drawn at random from a deck of cards. Find the probability of getting a queen.

**6)** A jar contains 3 red marbles, 7 green marbles and 10 white marbles. If a marble is drawn from the jar at random, what is the probability that this marble is white?

**7)** The blood groups of 200 people are distributed as follows: 50 have type A blood, 65 have type B, 70 have type O and 15 have type AB. If a person from this group is selected at random, what is the probability that this person has type O blood?

**8)** A die is rolled. Find the probability that the number obtained is greater than 4.

**9)** Two coins are tossed. Find the probability that one head only is obtained.

**10)** A die is rolled twice. Find the probability that the sum is equal to 5.

**11)** Two dice are rolled. Find the probability that the sum is

i. equal to 1;
ii. equal to 4;
iii. less than 7.

**12)** If a die is rolled once, what is the probability that

i. it will show an even number?
ii. it will show an odd number?
iii. it will show a prime number?
iv. it will show a multiple of 3?
v. it will show a multiple of 1?
vi. it will show a multiple of 7?

**13)** A deck of cards contains 52 cards, 13 from each suit. If a card is flipped over, what is the probability that it is a spade?

**14)** Three coins are tossed. What is the probability of getting
   i. all heads;
   ii. exactly one head;
   iii. at least one head?

**15)** A box contains 5 red, 3 white and 6 blue balls. If 3 balls are drawn at random, determine the probability that
   i. all 3 are blue;
   ii. 2 are red and 1 is white;
   iii. one of each colour is drawn.

**16)** Four dice are thrown simultaneously. Find the probability that all of them show the same face.